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Department of Economics Fall 2007

Econ 508

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Applied Econometrics

Problem Set 2

1. Introduction:

Data was read into R from 'gasq.data' and column items were redefined according to the following specifications:

```
> gas<-read.table("gasq.data",header =T)

> year<-gas$year

> Q<-gas$ln.y

> price<-gas$ln.p

> inc<-gas$ln.z
```

where,

Q: per capita personal gasoline consumption in thousands of gallons (at annual rates)

price: real price/ gallon of gasoline in 1982 $ (1 gallon = $ at 1982 prices)

inc: per capita personal income (in 1000's of 1982 $ at annual rates)

Note: Observations are quarterly observations from 1947.1 to 1997.1

```
> Q<-Q/4

> inc<-inc/4
```

The consumption and income data was adjusted for quarterly comparisons, where the simple nature of this adjustment owes itself to the fact that all variables in the dynamic model are in natural logarithms.

```
> Q<-ts(Q,start=1947,frequency=4)

> price<-ts(price,start=1947,frequency=4)

> inc<-ts(inc,start=1947,frequency=4)
```

The consumptions, income and price data was then redefined in a time-series format

for computational ease.

```
> LQ<-lag(Q,-1)

> DQ<-diff(Q)

> LDQ<-lag(DQ,-1)

> LLDQ<-lag(LDQ,-1)

> P<-price

> LP<-lag(P,-1)

> DP<-diff(P)

> LDP<-lag(DP,-1)

> LLDP<-lag(LDP,-1)

> I<-inc

> LI<-lag(I,-1)

> DI<-diff(I)

> LDI<-lag(DI,-1)

> LLDI<-lag(LDI,-1)
```

Lagged terms were constructed from consumption, income and price data respectively, followed by difference terms based on the current and lagged observations.

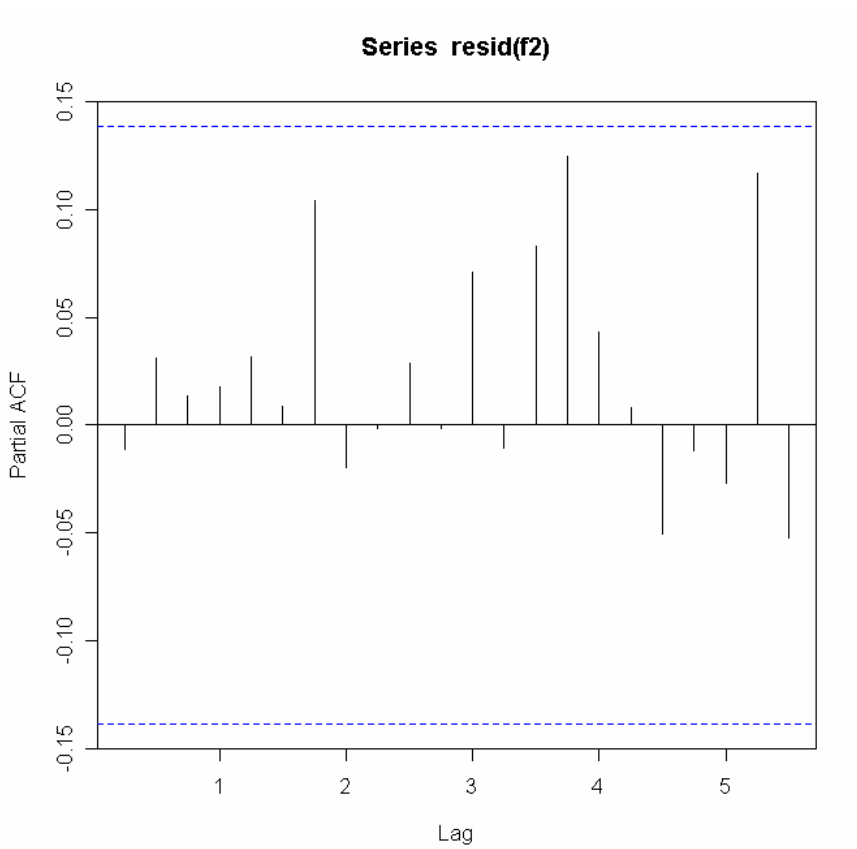
Original model (expanded version)

> f2<-dyn$lm(Q~LQ+LDQ+P+DP+LDP+I+DI+LDI)

> acf(resid(f2))

The residuals do not show significant AR behavior, so AR behavior is taken by the model. They also do not show significant MA behavior as the following test shows:

> pacf(resid(f2))



So this is a proper model if we accept the hypothesis that there is no heteroskedasticity in the model. The information criteria for this model shows:

AIC(f2, k=log(length(fitted(f2)))) # = -1580.337

If we take a summary of the results:

	Coefficients	**Estimate Std. Error**	**t value**	**Pr(>t)**
(Intercept)	-0.090280	0.021608	-4.178	4.48e-05
LQ	0.934802	0.015812	59.118	< 2e-16
LDQ	-0.074536	0.069097	-1.079	0.282
P	-0.008649	0.001934	-4.472	1.33e-05
DP	-0.259506	0.008207	-31.621	< 2e-16
LDP	-0.015496	0.020549	-0.754	0.452
I	0.126619	0.029222	4.333	2.38e-05
DI	-0.165425	0.122978	-1.345	0.180
LDI	0.045057	0.120121	0.375	0.708
Residual Standard error	0.004089 on 190 df			
Multiple R-Squared	0.9976	Adjusted R-Squared	0.9975	
F-Statistic	1.004e+04 on 8 and 190 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-0.0220141	-0.0023730	0.0001413	0.0021832	0.0157879

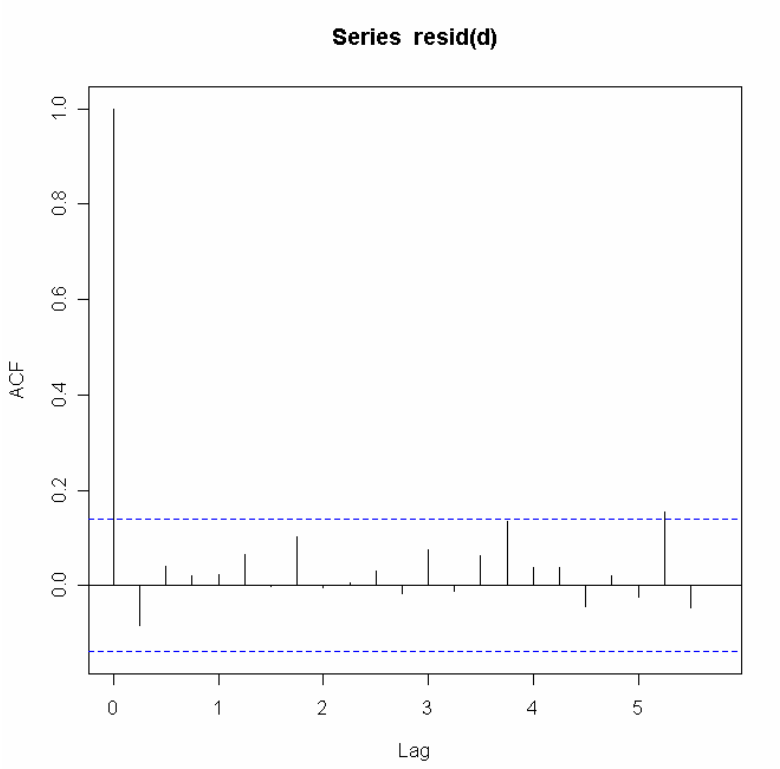
The results in the above table show insignificant coefficient values for LDQ, LDP, DI, and LDI. Hence, we drop these terms in the second estimation:

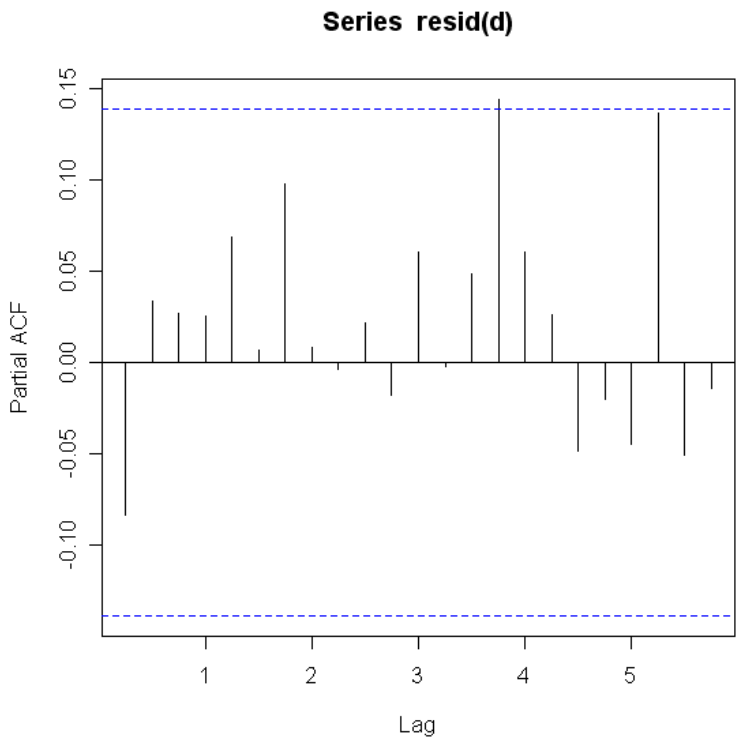
d<-dyn$lm(Q~LQ+P+DP+I)

acf(resid(d))

pacf(resid(d))

The residuals in this model show some minor AR behavior; furthermore, there is significant evidence of MA behavior, as shown in the plots on the next page.


Series resid(d)
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
Lag


Series resid(d)
Partial ACF
0.15
0.10
0.05
0.00
-0.05
-0.10
1
2
3
4
5
Lag

The model exhibits a modest value based on the SIC Criteria, given by:

> AIC(d,k=log(length(fitted(d)))) = -1605.054.

At the same time, the statistical significance of all coefficients in the regression summary should be noted:

	Coefficients	**Estimate Std. Error**	**t value**	**Pr(>t)**
(Intercept)	-0.089299	0.020312	-4.396	1.81e-05
LQ	0.934210	0.014857	62.879	< 2e-16
P	-0.008308	0.001798	-4.622	6.91e-06
DP	-0.255420	0.007354	-34.732	< 2e-16
I	0.124480	0.027371	4.548	9.51e-06
Residual Standard error	0.004093 on 195 df			
Multiple R-Squared	0.9976	Adjusted R-Squared	0.9976	
F-Statistic	2.042e+04 on 4 and 195 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-2.139e-02	2.503e-03	2.335e-05	2.099e-03	1.647e-02

Furthermore, the results help affirm the following:

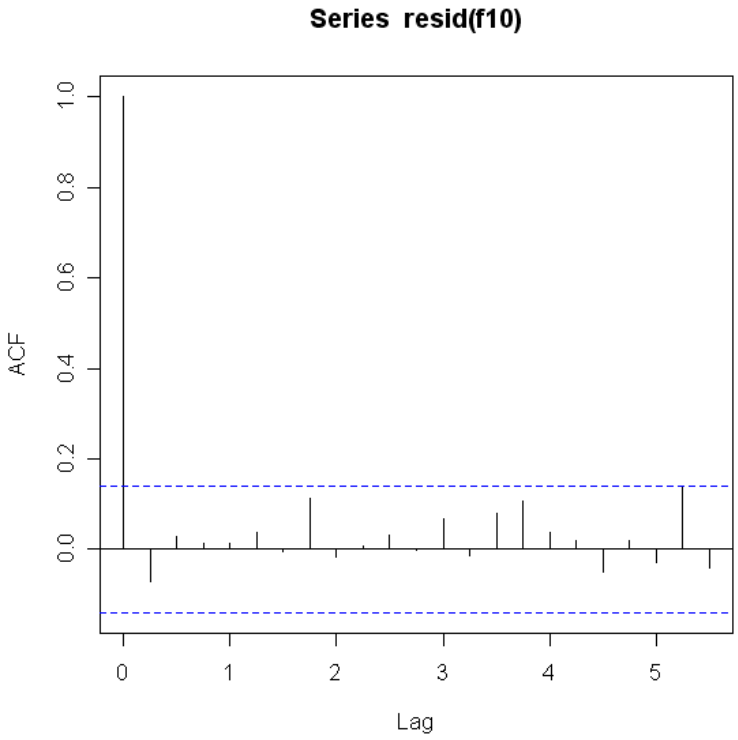
- **the coefficient of lagged consumption is positive, which is consistent with the 'sticky' behavior of consumption,**
- **the coefficient of price is negative, which is consistent with demand behavior,**
- **the coefficient of price difference is also positive, which is consistent with more adjustment in the long run,**
- **the coefficient of income is positive, which is consistent with the theory of a normal good,**
- **the negative intercept is also appropriate, because values are in log form**

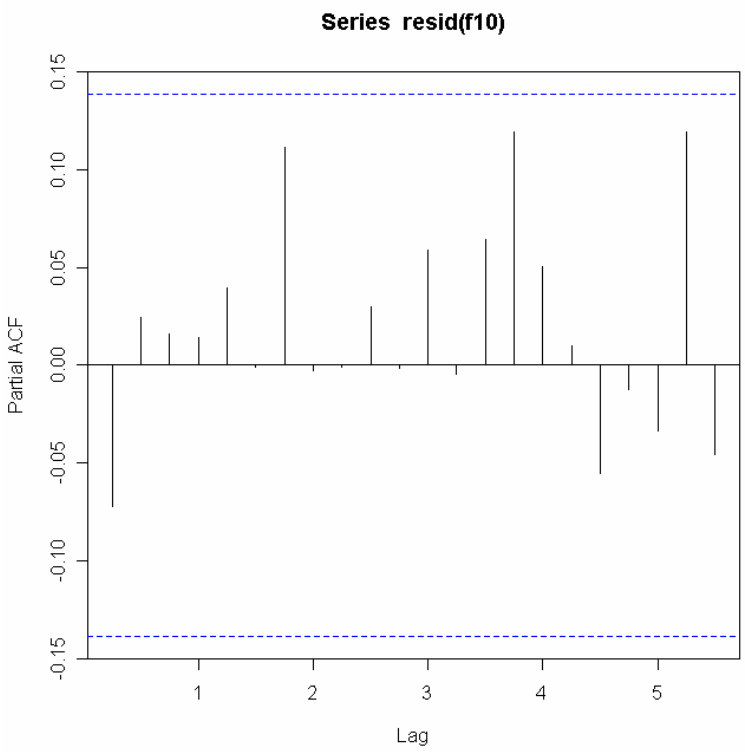
If the presence of moderate AR and MA behavior is a cause for concern, the closest model to the current one, in terms of information criteria, is the following:

```
> f10<-dyn$lm(Q~LQ+LDQ+P+DP+I)

> acf(resid(f10))

> pacf(resid(f10))
```


Series resid(f10)
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
Lag


Series resid(f10)
Partial ACF
0.15
0.10
0.05
0.00
-0.05
-0.10
-0.15
1
2
3
4
5
Lag

Both plots show completely insignificant levels of serial correlation between the residuals, and the SIC information criterion is also not much higher than the previous (d) model:

> AIC(f10, k=log(length(fitted(f10)))) = -1593.622.

However, the regression results present a different picture:

	Coefficients	**Estimate Std. Error**	**t value**	**Pr(>t)**
(Intercept)	-0.084182	0.020720	-4.063	7.05e-05
LQ	0.938934	0.015274	61.471	< 2e-16
LDQ	-0.028324	0.027152	-1.043	0.298
P	-0.007988	0.001812	-4.409	1.72e-05
DP	-0.258720	0.007757	-33.352	< 2e-16
I	0.117926	0.027883	4.229	3.62e-05
Residual Standard error	0.004083 on 193 df			
Multiple R-Squared	0.9976	Adjusted R-Squared	0.9976	
F-Statistic	1.611e+04 on 5 and 193 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-2.140e-02	-2.424e-03	-4.708e-06	2.103e-03	1.646e-02

As the table shows, the inclusion of the extra variable does not yield much as it is statistically insignificant, while the other variables share the same coefficients as their counterparts in the previous regression model. As a result, the previous model is retained as the model of choice for this analysis.

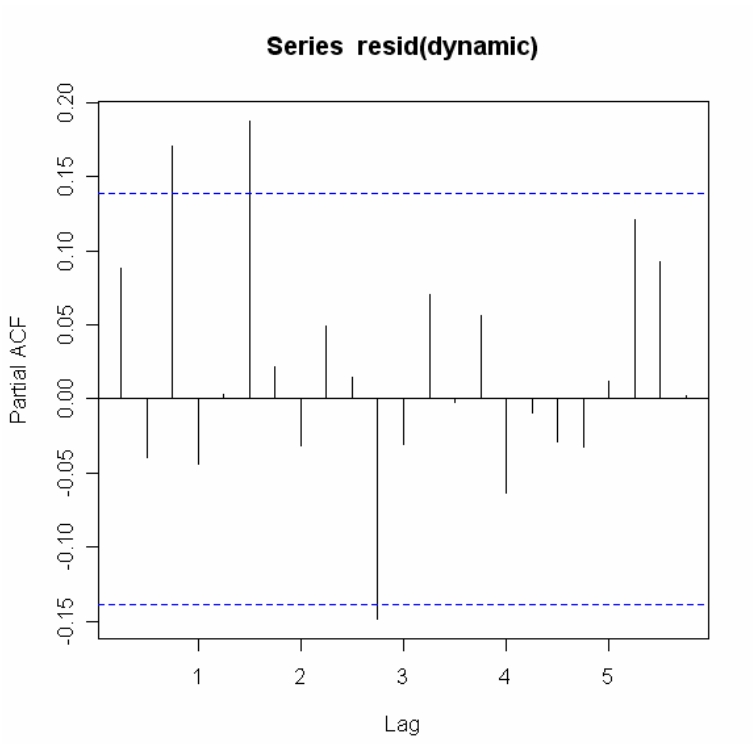
2.Long run income and price elasticities:

From question one, we found the following regression the best one among based on the SIC information criteria, which doesn't show significant serial correlations:

> dynamic<-dyn$lm(Q~LQ+P+DP+I)

	Coefficients	Estimate Std. Error	t value	Pr(>t)
(Intercept)	-0.357197	0.081248	-4.396	1.81e-05
LQ	**0.934210**	0.014857	62.879	< 2e-16
P	**-0.033232**	0.007191	-4.622	6.91e-06
DP	**-1.021679**	0.029416	-34.732	< 2e-16
I	**0.124480**	0.027371	4.548	9.51e-06
Residual Standard error	0.004093 on 195 df			
Multiple R-Squared	0.9976	Adjusted R-Squared	0.9976	
F-Statistic	2.042e+04 on 4 and 195 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-0.0855445	-0.0100123	0.0000934	0.0083950	0.0658905

The results of this model are valid in the sense that there no significant serial correlation among the residuals:



Among the other models, this model has much a better plot for serial correlation among residuals than the other ones.

For the long run version, the difference variables should be set to zero and all the lagged variables should be assigned the same value, therefore:

$$Qeq = -5.429 - 0.505\ Peq + 1.892\ Ieq$$

where Qeq= Equilibrium consumption

Peq= Equilibrium price

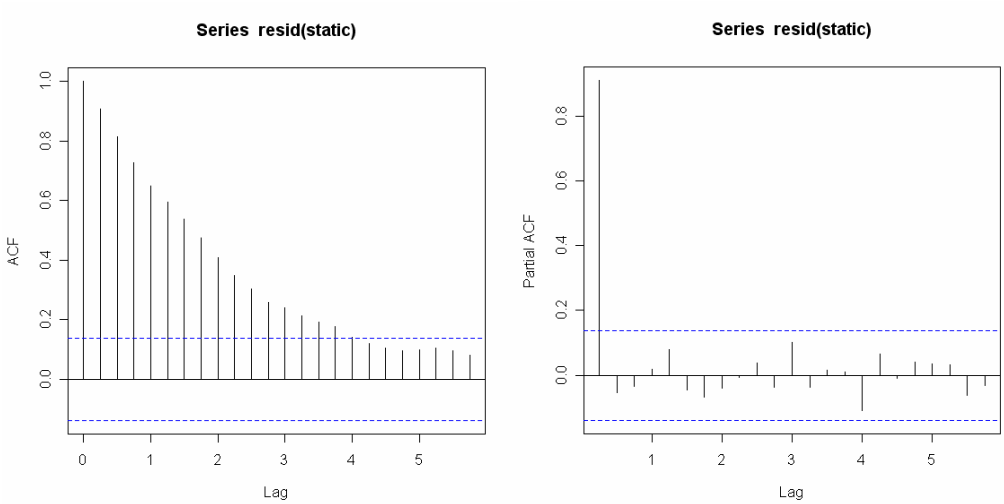
Ieq=Equilibrium income

The simple static model is estimated as:

> static<-dyn$lm(Q~P+I)

	Coefficients	**Estimate Std. Error**	**t value**	**Pr(>t)**
(Intercept)	-5.26650	0.11217	-46.95	<2e-16
P	-0.40753	0.01964	-20.75	<2e-16
I	1.75935	0.04194	41.95	<2e-16
Residual Standard error	0.07936 on 198 df			
Multiple R-Squared	0.9442	Adjusted R-Squared	0.9436	
F-Statistic	1674 on 2 and 198 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-0.217743	-0.042519	0.006042	0.061758	0.170490

The static model is spurious because of the high serial correlations among the residuals of the model:



The log form in which the data is considered allows the regression coefficients to be directly interpreted as elasticities. The long-run income and price elasticities are computed as (using the data in the summary (longrun)):

Long-run Price Elasticity:	-0.5
Long-run Income Elasticity:	+1.89

The static model elasticities are then compared to the long-run values,

Static Model Price Elasticity:	-0.4
Static Model Income Elasticity:	+1.759

The problem with the static model is the excessive amount of serial correlation in the residuals, which invalidates its results. As a result, we cannot accept the static model.

Qeq = -5.429 – 0.505 Peq + 1.892 Ieq

Qs = -5.266 – 0.408 Ps + 1.759 Is

Although the static model's results are totally invalid and biased in general, they look pretty close to the equilibrium results taken from the dynamic model.

The difference between the results from the two models can be explained by the biasness in regression owing to serial correlation between residuals.

Yet, computation of the exact confidence intervals is required to formalize this argument. The correct way for checking would be to find the confidence intervals for the coefficients in the long run model and then check to see if the coefficients of the static model lie in that region. This is the most preferred approach to adopt, as test statistics cannot be run for the spurious static model.

In order to do this one should consider that the coefficients of the equilibrium model have complicated distributions, and either the Delta-Method or the Bootstrap technique should be used to calculate their confidence regions:

```
DP<-diff(P)
DP<-c(DP,DP[200])
test<-dyn$lm(Q~LQ+P+DP+I)
uhat<-test$resid
h1<-rep(0,1000)
h2<-rep(0,1000)
h3<-rep(0,1000)

for(i in 1:1000)
{
temp<-test$fit+sample(uhat,replace=TRUE)
temp<-c(temp,temp[200])    #temp<-ts(temp,start=1947,frequency=4)!!!
b<-coef(dyn$lm(temp~LQ+P+DP+I))
h1[i]<-b[3]/(1-b[2])
h2[i]<-b[5]/(1-b[2])
h3[i]<-b[2]
```

```
}
quantile(h1,c(0.025,0.975))
quantile(h2,c(0.025,0.975))
```

Price elasticity confidence interval: **– 0.408 ε (-0.6166335, -0.3183230)**

Income elasticity conf interval: **+ 1.759 ε (1.527369, 2.107277)**

Revenue Implications

As for the tax revenue implications of the calculated long-run elasticties, it can be said that as the price elasticity is -0.5, or fairly inelastic, a tax-induced price rise of gasoline will increase the revenue accumulated.

- It is assumed that the elasticities will remain constant at different tax rates.
- Also the demand is taken as the dimand in the final date:

$$D= \exp(0.6302744) = 1.878126$$

Using the simple rule for differentiating products of variables and first order Taylor simplifications:

$$\Delta r=D*\Delta P*(1+e)*1000$$

The exact net capita revenue gain from the imposition of an additional tax, given the current tax rate of 14 cents per gallon, is as follows:

For an additional 10 cent per gasoline tax:

$$\Delta r=92.97\$$$

For an additional 20 cent per gasoline tax,

$$\Delta r= 185.94\$$$

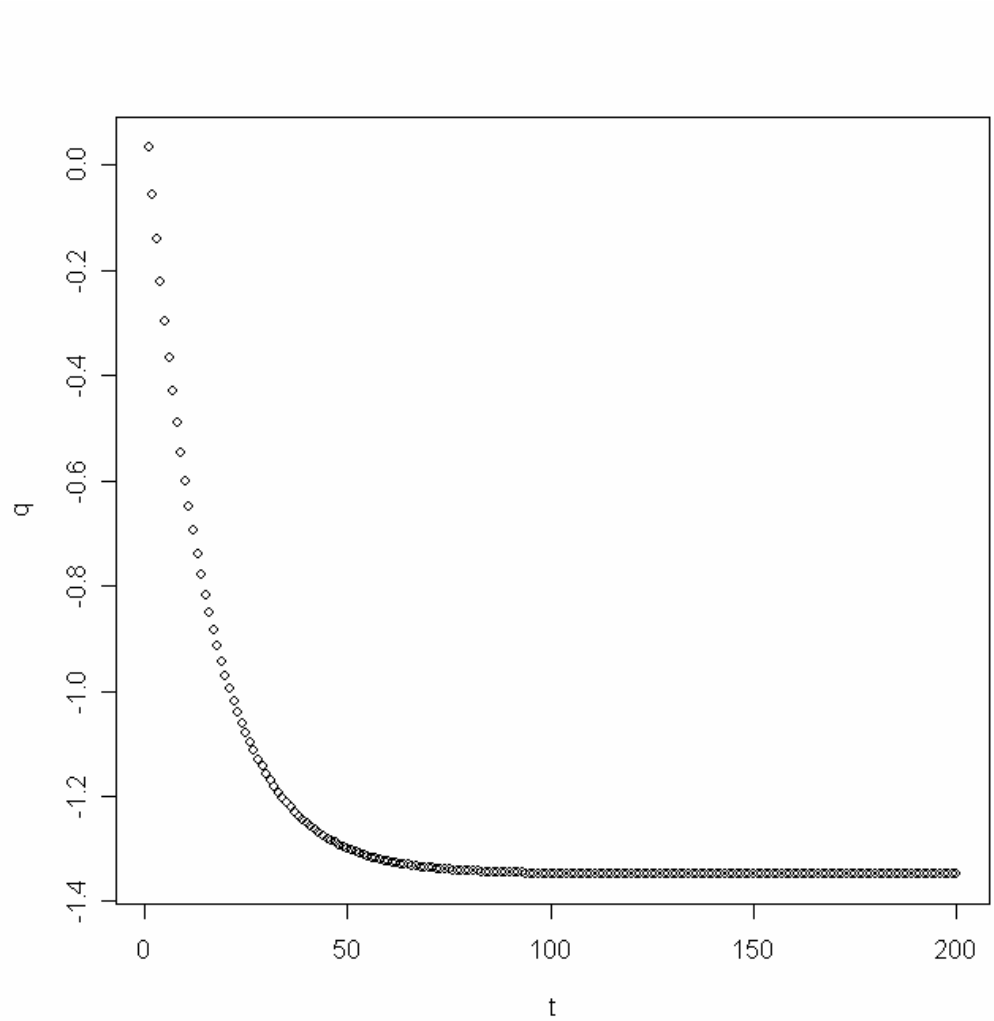
For an additional 50 cent per gasoline tax,

$$\Delta r= 464.85\$$$

3.Income Impulse Response function:

```
> q = NULL; t = NULL;p = 0;Lp = 0;Ld = 0;inc = 1
 > for (i in 1:200){
+ t = c(t,i)
+ d = -0.089 + 0.934 * Ld - 0.008 * p - 0.255 * (p - Lp) + 0.124 * inc
+ q = c(q,d)
+ inc = 0
+ Ld = d
 + }
```

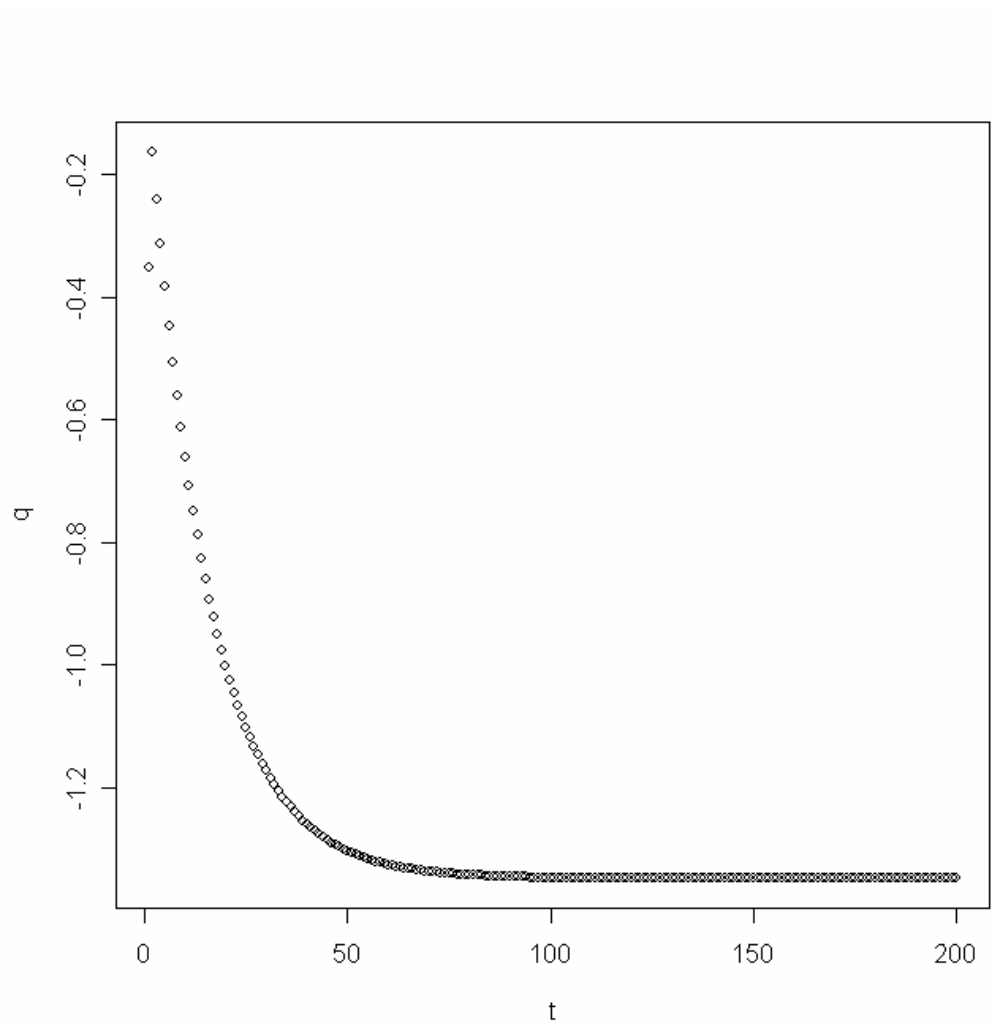
The non-cumulative income impulse response function for the final model is presented below:



Price Impulse Response function:

```
> q = NULL;t = NULL;p = 1;Lp = 0;Ld = 0;inc = 0
> for(i in 1:200){
+ t = c(t,i)
+ d = -0.089 + 0.934 * Ld - 0.008 * p - 0.255 * (p- Lp) + 0.124 * inc
+ q = c(q,d)
+ inc = 0
+ Ld = d
+ Lp = p
 + p = 0  }
```

The non-cumulative price impulse response function for the final model is presented below:



Error Correction form:

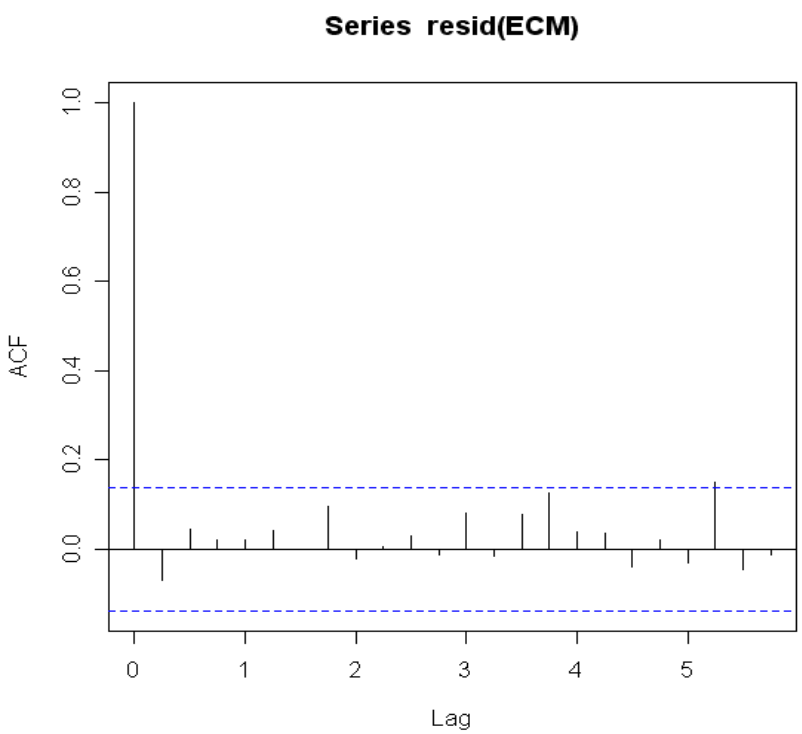
To write the model in the Error Correction form, we first run the following regression, in which the coefficients of the long term model are hidden in the coefficients of the **Intercept, LQ, LP, LI;**

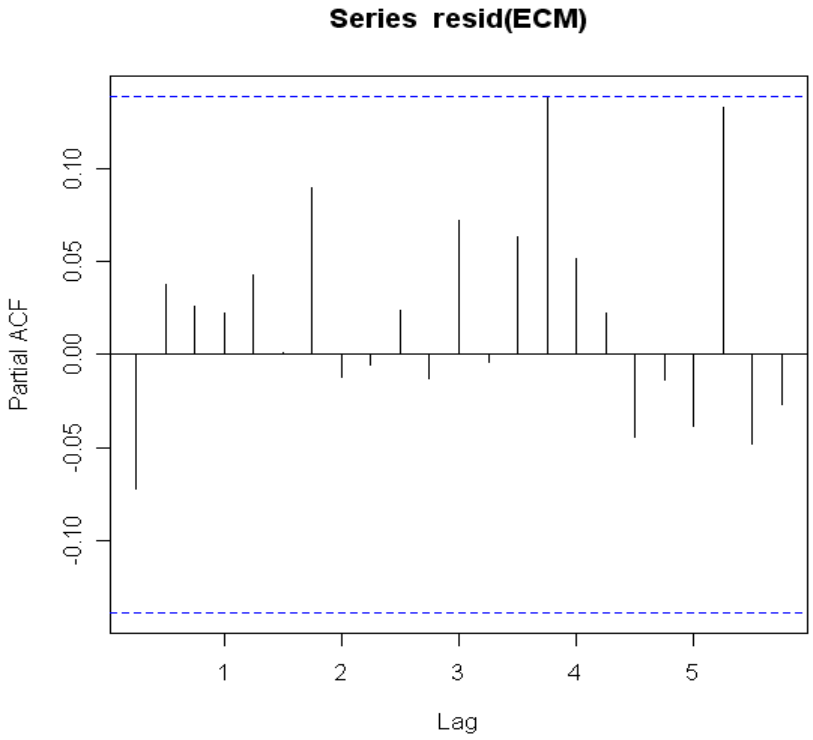
As shown by the acf plot below, the error structure of the model shows no serial correlation behavior, and we can assume that it is valid:

```
> ECM<-dyn$lm(DQ~DP+DI+LQ+LP+LI)

> acf(resid(ECM2))

> pacf(resid(ECM))
```





The residuals show no significant correlation behavior, so we can trust the results within the confidence interval.

> summary(ECM)

	Coefficients	**Estimate Std. Error**	**t value**	**Pr(>t)**
(Intercept)	-0.092749	0.020322	-4.564	8.90e-06
DP	-0.267658	0.007575	-35.337	< 2e-16
DI	**-0.068111**	**0.117930**	**-0.578**	**0.564**
LQ	-0.067237	0.014814	-4.539	9.91e-06
LP	-0.008821	0.001815	-4.859	2.42e-06
LI	0.129634	0.027417	4.728	4.35e-06
Residual Standard error	0.004074 on 194 df			
Multiple R-Squared	0.877	Adjusted R-Squared	0.8739	
F-Statistic	276.7 on 5 and 194 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-2.203e-02	2.443e-03	5.285e-05	2.168e-03	1.602e-02

The regression results for the Error Correction Model present a statistically insignificant coefficient for the DI variable in the model.

As a result, a second Error Correction Model was estimated, excluding the DI, or difference in per capita personal income term. (This model will be presented at the

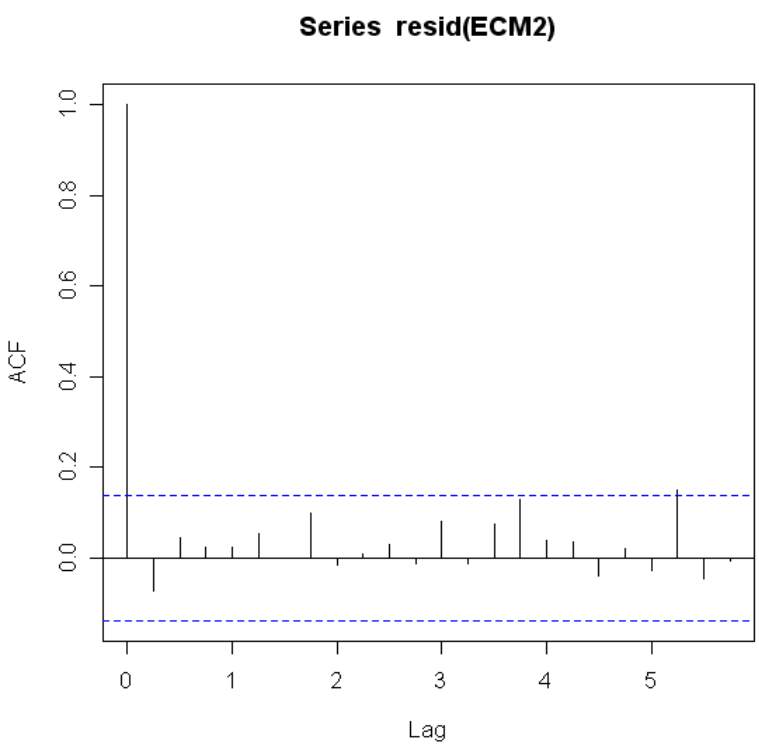
end of this section; before doing so, the previous regression model will be redefined in the ECM form)

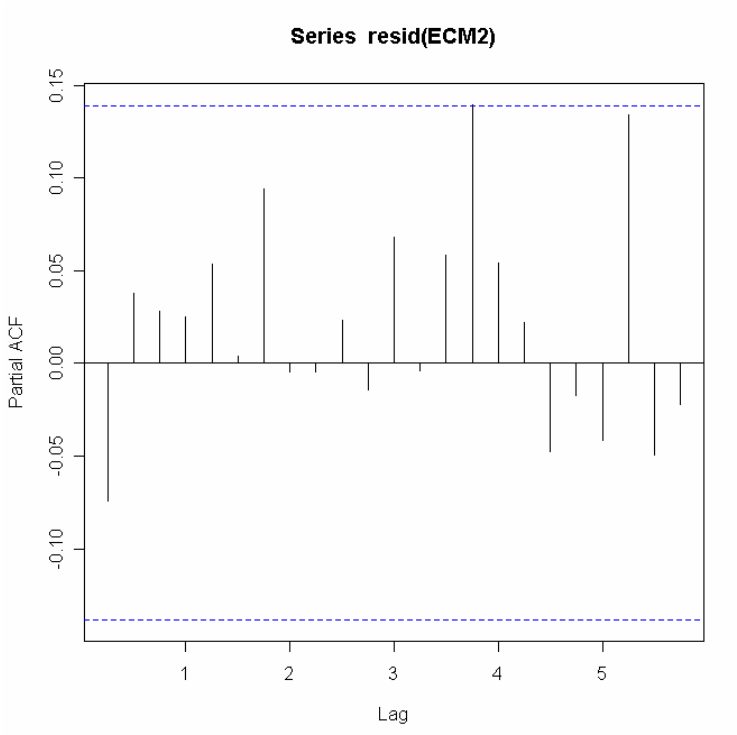
$$\Delta y_t = -0.268\Delta p_t - 0.068\Delta I_t - 0.067\left[\Delta y_{t-1} + 0.131 p_{t-1} - 1.928 I_{t-1}\right]$$

The coefficient of ΔI looks unreasonable, (It should be positive, but is in fact negative); this can be attributed to the insignificance of this variable. As previously mentioned, and in light of this result, a new model will be constructed by dropping the ΔI term in the current model:

```
> ECM2<-dyn$lm(DQ~DP+LQ+LP+LI)

> acf(resid(ECM2))
```





The partial autocorrelation function versus lag plot shows that there is only a minor occurrence of serial correlation in the residuals.

The Error Correction Model presents a framework within which changes in the quantity of gasoline consumed between periods can be decomposed into the following:

- Shocks from disequilibrium in the model owing to past disturbances, which impose themselves in the ECM based on a 'sensitivity' or 'speed of adjustment' term,
- Shocks emerging in the current period of time.

4.Implications for gasoline demand from alternative model

The price elasticity of demand is expressed as:

$$\eta = \frac{\partial y}{\partial x} = \beta_2 + 2\beta_3 p_t + \beta_4 z_t$$

(i) The implication that gasoline demand is more elastic as price increases is plausible, as an increase in the price of gasoline will make consumers of gasoline more sensitive to the impact of gasoline on their respective budgets. This is primarily due to the larger share of personal income spent on gasoline; hence, a change in the price of gasoline will significantly impact the total expenditure of a consumer and induce him or her to alter gasoline consumption correspondingly.

(ii) The implication that gasoline demand is less elastic as income increases is also plausible, as it relies on the share of total personal income devoted to expenditure on gasoline. The larger the magnitude of personal income, the smaller the share of gasoline expenditure in the budget; as a result, a change in the price of gasoline will not induce a significant change in total expenditure, and will not promote a consumer to significantly the consumption of gasoline.

Now, revenue is maximized when $\eta = -1$; per capita income is x_0

In order to compute the price at which gasoline revenue is maximized, assuming the alternative model;

$$\beta_2 + 2\beta_3 p_t + \beta_4 z_t = -1$$

And $z_t = x_o$,

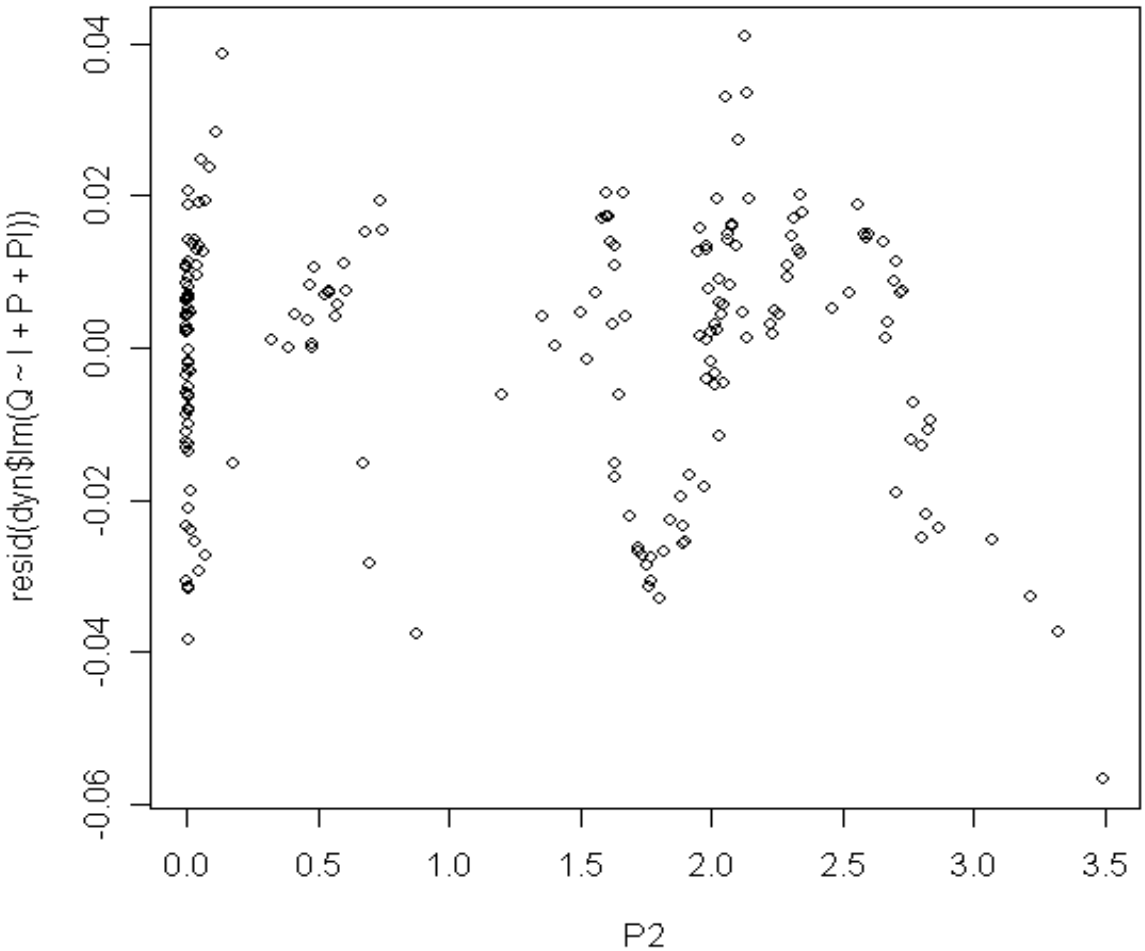
As a result, $$p^* = \left(\frac{-1 - \beta_4 x_0 - \beta_2}{2\beta_3} \right)$$

For estimating the partial plot,

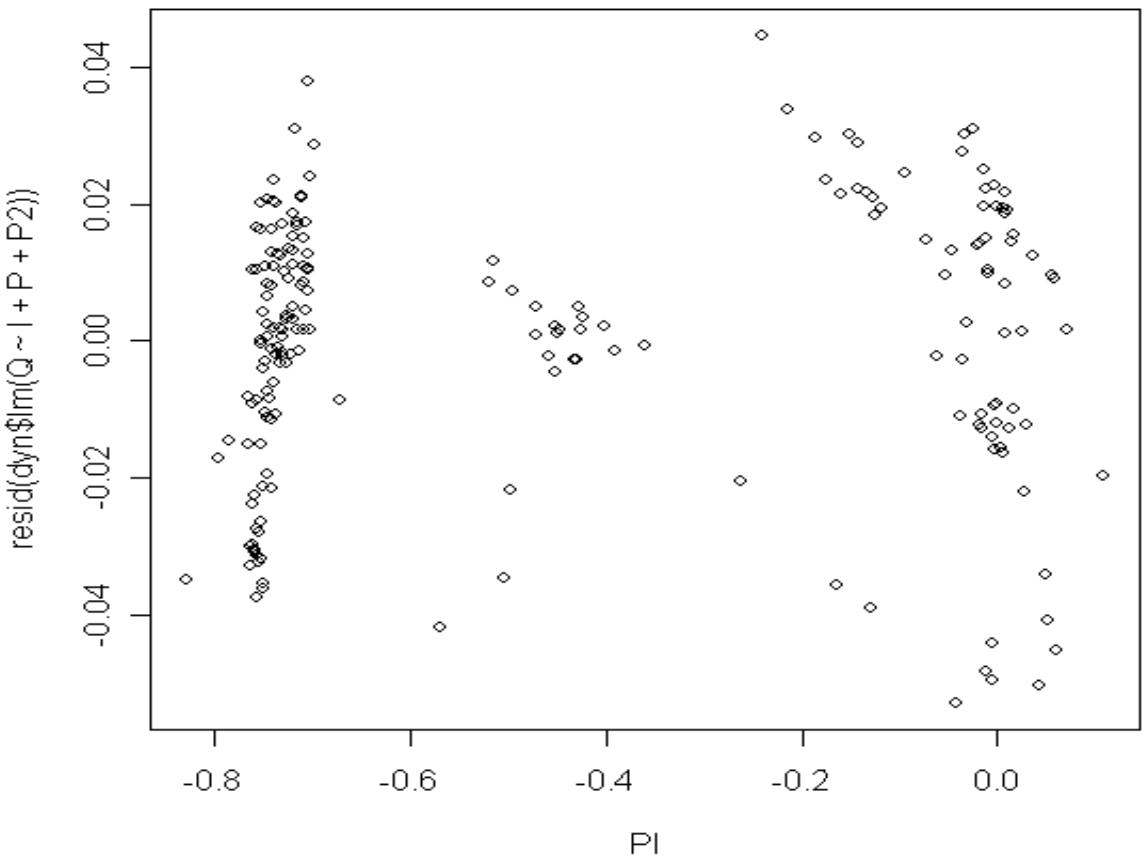
```
> P2<-NULL
> PI<-NULL
> for(i in 1:length(P))
+ {
+ P2<-c(P2,P[i]*P[i])
+ PI<-c(PI,P[i]*I[i])
+ }
```

The partial residual plots, graphically illustrating the negative correlation between the residuals and PI, P2 respectively, are estimated:

```
> plot(P2,resid(dyn$lm(Q~I+P+PI)))
```



```
> plot(PI,resid(dyn$lm(Q~I+P+P2)))
```



While the plots emphasize the strong negative correlation between the residuals of the models and PI, P2 respectively, a formal test for the significance of these effects is required at this stage:

```
> x<-resid(dyn$lm(Q~I+P+PI))

> y<-resid(dyn$lm(Q~I+P+P2))
```

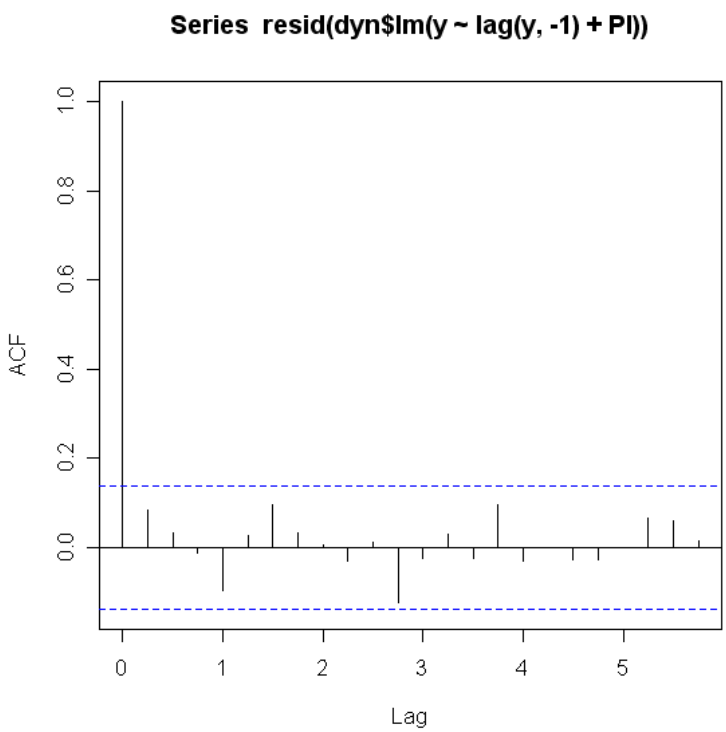
For PI term:

> summary(dyn$lm(y~lag(y,-1)+PI))

	Coefficients	Estimate Std. Error	t value	Pr(>t)
(Intercept)	0.0001023	0.0009914	0.103	0.918
DQ	0.9009283	0.0296244	30.412	<2e-16
P1	-0.0001752	0.0017661	-0.099	0.921
Residual Standard error	0.008222 on 197 df			
Multiple R-Squared	0.8245	Adjusted R-Squared	0.8245	
F-Statistic	462.8 on 2 and 197 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-0.0343077	-0.0040477	-0.0003652	0.0040227	0.0278369

, and with an acceptable ACF:

> acf(resid(dyn$lm(y~lag(y,-1)+PI)))



And for the P2 term:

> summary(dyn$lm(x~lag(x,-1)+P2))

	Coefficients	Estimate Std. Error	t value	Pr(>t)
(Intercept)	-0.0003305	0.0008464	-0.391	0.697
DX	0.8546223	0.0334378	25.559	<2e-16
P2	0.0004445	0.0005322	0.835	0.405
Residual Standard error	0.007874 on 197 df			
Multiple R-Squared	0.7683	Adjusted R-Squared	0.766	
F-Statistic	326.7 on 2 and 197 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-0.0322351	-0.0041486	-0.0006364	0.0040376	0.0284184

, with an acceptable ACF:

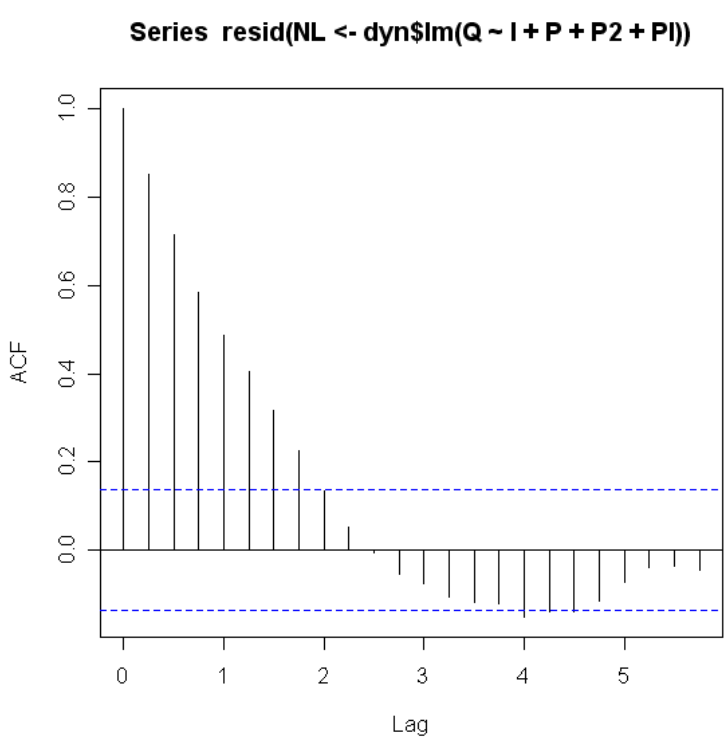
> acf(resid(dyn$lm(x~lag(x,-1)+P2)))

As both regression summaries show, the coefficients of PI, P2 respectively, are statistically insignificant.

d) The non constant elasticity model is,

>NL<-dyn$lm(Q~I+P+P2+P1)

> acf(resid(NL<-dyn$lm(Q~I+P+P2+P1)))



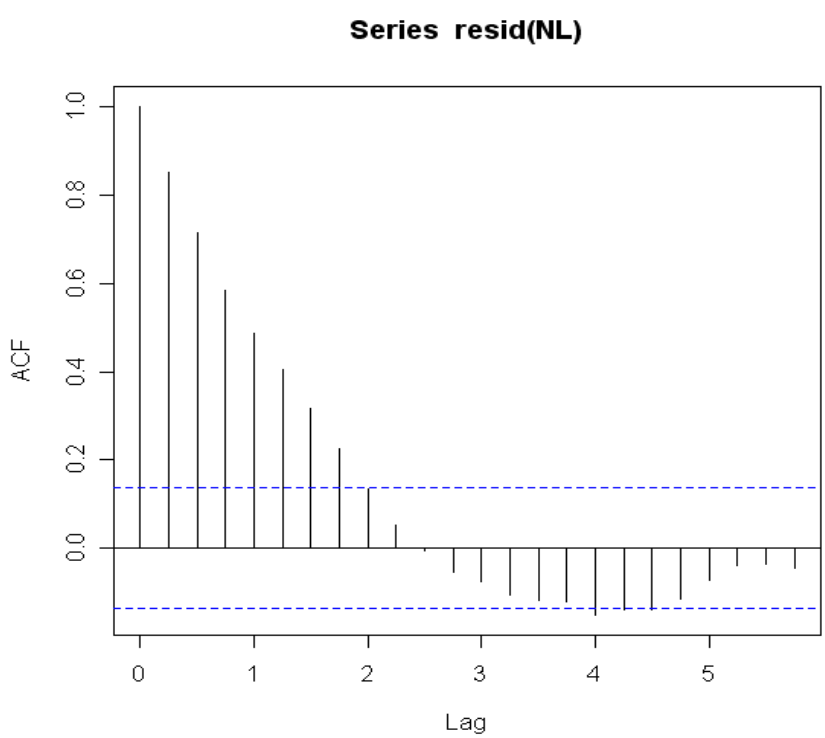
Neglecting these problems in this current question (and addressing them in question five) we will take a look at the calculated coefficients:

> summary(NL)

	Coefficients	**Estimate Std. Error**	**t value**	**Pr(>t)**
(Intercept)	-1.847868	0.050803	-36.373	< 2e-16
I	2.571179	0.077643	33.115	< 2e-16
P	-0.643421	0.046452	-13.851	< 2e-16
P2	-0.067818	0.009384	-7.227	1.08e-11
P1	0.717018	0.059308	12.090	< 2e-16
Residual Standard error	0.01501 on 196 df			
Multiple R-Squared	0.9684	Adjusted R-Squared	0.9677	
F-Statistic	1499 on 4 and 196 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-0.046363	-0.009862	0.001787	0.009940	0.034764

All coefficients, including PI and P2, seem plausibly significant, in contrast to when they were separately tested; owing to the spurious regression we are running, the existence of highly serially correlated coefficients is no surprise.

> acf(resid(NL))



Although the regression is spurious, it justifies part (a) for the sign of the coefficients. The estimate of the optimal revenue maximizing price level is:

```
> b<-lm(Q~I+P+P2+PI)$coef

> -(1+b[3]+b[5]*log(15))/(2*b[4])
```

Taking the exponent of the result,

```
>exp(0.6783594)
```

, the point estimate is **1.822119**

We write the following code for the Bootstrap Method to find the confidence interval:

```
> test<-lm(Q~I+P+P2+PI)

> uhat<-test$resid

> plot(uhat)

> acf(uhat)

> h<-rep(0,1000)

> plot(h)

> h[100]

> for(i in 1:1000)

+ {

+ gash<-test$fit+sample(uhat,replace=TRUE)

+ b<-lm(gash~I+P+P2+PI)$coef

+ h[i]<-(-(1+b[3]+b[5]*log(15))/(2*b[4]))

+ }

> plot(h)

> quantile(h,c(0.025,0.975))
```

And the values are:

2.5%	97.5%
1.521689	2.886452

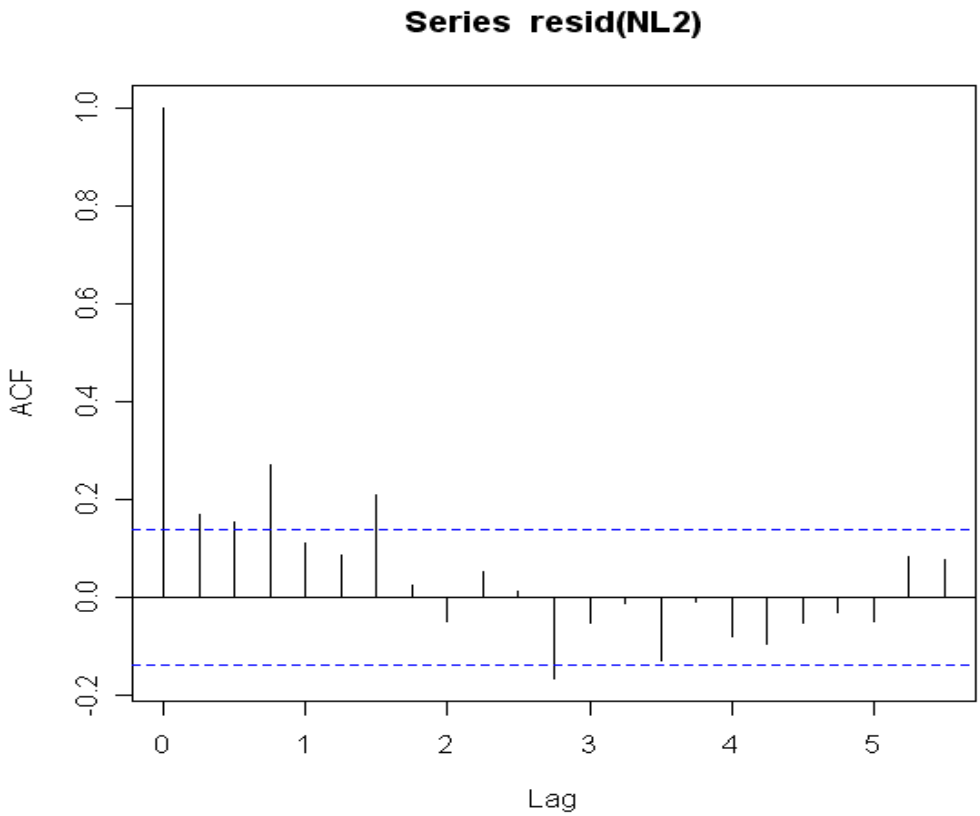
5.Final specification of the model (departure from Gaussian linear model conditions):

As done earlier (in Q.2), the equilibrium coefficients here are a scaled version of the static model. Estimating the dynamic model we have:

```
> NL2<-dyn$lm(Q~LQ+LDQ+I+P+P2+PI)

> acf(resid(NL2))
```

Although there is improvement in the correlation between residuals, nevertheless they are still significant.



> summary(NL2)

	Coefficients	Estimate Std. Error	t value	Pr(>t)
(Intercept)	-0.674054	0.096318	-6.998	4.22e-11
LQ	0.637329	0.048573	13.121	< 2e-16
LDQ	0.288009	0.063707	4.521	1.08e-05
I	0.939072	0.135808	6.915	6.77e-11
P	-0.224343	0.044189	-5.077	9.04e-07
P2	-0.020069	0.007194	-2.790	0.00581
P1	0.253181	0.053223	4.757	3.85e-06
Residual Standard error	0.01004 on 192 df			
Multiple R-Squared	0.9856	Adjusted R-Squared	0.9852	
F-Statistic	2196 on 6 and 192 df	p-value < 2.2e-16		
Residuals				
Min	IQ	Median	3Q	Max
-0.0507320	-0.0057364	0.0001465	0.0057099	0.0288740

Calculating the equilibrium model we will have:

Qeq = -1.856 + 2.589Ieq - 0.618Peq - 0.055Peq^2 + 0.698 PeqIeq

Comparing above with the static model:

Q = + 2.571I +-0.643P – 0.068P^2 – 0.717PI

Although the values seem similar, it is better to run a test to see if they are significantly different or not; once again, given the spurious nature of the static model, it is preferred to run the test for the dynamic model. The existence of non linearity means that the bootstrap or Delta-Method is required to determine the confidence intervals.

For the sake of departure, we have already shown that there is some level of serial correlation between the error terms; it is also possible to run this test for heteroskedasticity.

The LM Test is applied as follows:

```
> resid<-resid(NL2)

> var=sum(resid^2)

> g<-(resid^2)/var

> OLS<-dyn$lm(g~P+I+Q)

> resid<-resid(OLS)

> LM<-sum(resid^2)/2
```

We can treat this LM value of 0.01444683 as ÷2, and comparing to a table of values it is obvious that there is no statistical significance; hence, hetroskedasticity is rejected

Appendix

First experiment: Reduction in original dynamic model by dropping the LDI term

The next step was to drop the lagged variable, LDI, or the lagged difference for per capita personal income, and run the same test for the new albeit reduced model:

```
> f3<-dyn$lm(Q~LQ+LDQ+P+DP+LDP+I+DI)
> acf(resid(f3))
> pacf(resid(f3))
> AIC(f3, k=log(length(fitted(f3))))
```

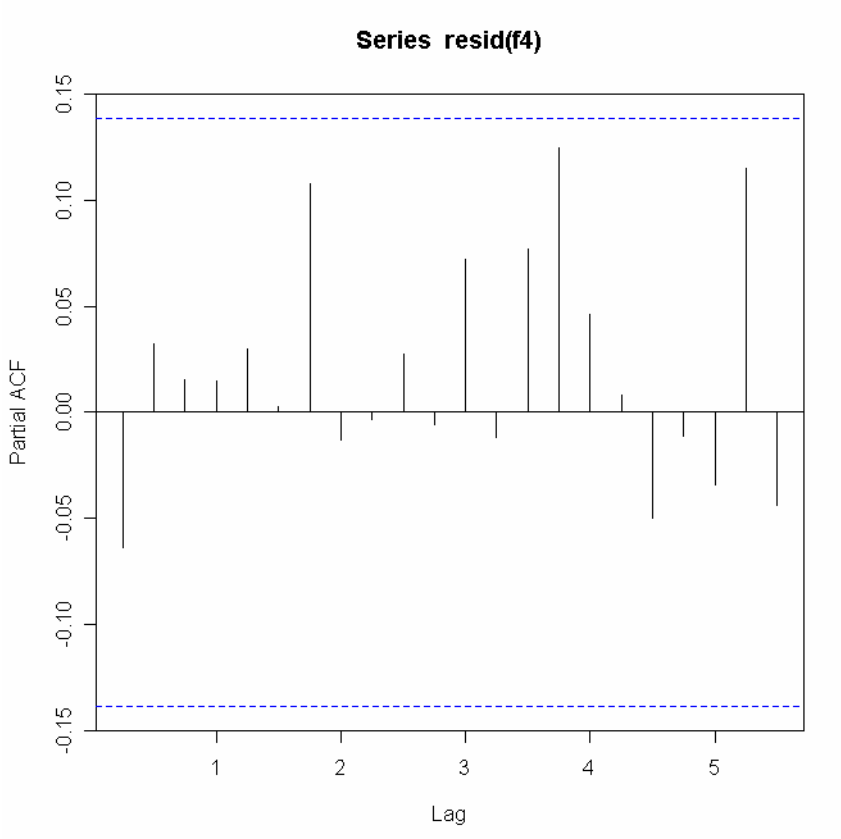
As the plot shows, there was no significant residual autocorrelation for any of the lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1585.483.

Second experiment: Reduction in original dynamic model by dropping the LDP term

The next step was to drop the lagged variable, LDP, or the lagged difference for real price per gallon of gasoline, and run the same test for the new reduced model:

```
> f4<-dyn$lm(Q~LQ+LDQ+P+DP+I+DI+LDI)
> acf(resid(f4))
> pacf(resid(f4))
> AIC(f4, k=log(length(fitted(f4))))
```



As the plot shows, there was no significant residual autocorrelation for any of the lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1585.036.

<u>Third experiment: Reduction in original dynamic model by dropping the LDQ and LDP terms</u>

The next step was to drop the lagged variable, LDQ, or the lagged difference for per capita personal consumption on gasoline, and the lagged variable, LDP, or the lagged difference for real price per gallon of gasoline, and run the same test for the new reduced model:

```
> f5<-dyn$lm(Q~LQ+P+DP+I+DI+LDI)
> acf(resid(f5))
> pacf(resid(f5))
> AIC(f5, k=log(length(fitted(f5))))
```

As the plot shows, there was no significant residual autocorrelation for any of the lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1589.372.

Fourth experiment: Reduction in original dynamic model by dropping the LDI and LDP terms

The next step was to drop the lagged variable, LDI, or the lagged difference for per capita personal income, and the lagged variable, LDP, or the lagged difference for real price per gallon of gasoline and run the same test for the new reduced model:

```
> f6<-dyn$lm(Q~LQ+LDQ+P+DP+I+DI)
> acf(resid(f6))
> pacf(resid(f6))
> AIC(f6, k=log(length(fitted(f6))))
```

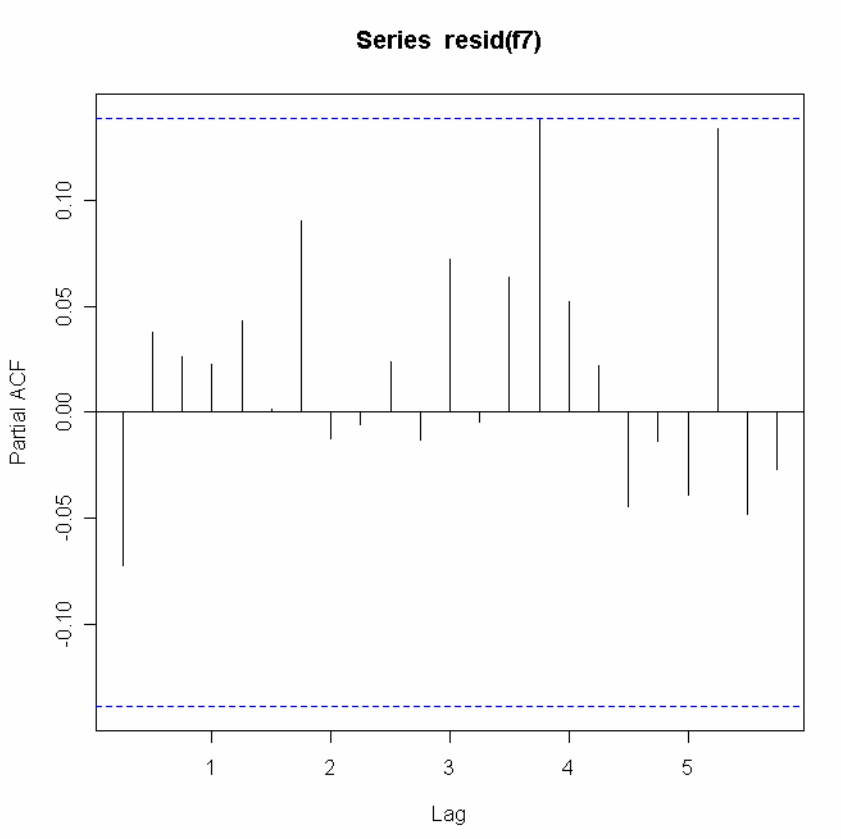
As the plot shows, there was no significant residual autocorrelation for any of the lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1590.017.

Fifth experiment: Reduction in original dynamic model by dropping the LDQ, LDI and LDP terms

The next step was to drop the LDQ, or the lagged difference for per capita personal consumption on gasoline lagged variable, LDI, or the lagged difference for per capita personal income, and the lagged variable, LDP, or the lagged difference for real price per gallon of gasoline and run the same test for the new reduced model:

```
> f7<-dyn$lm(Q~LQ+P+DP+I+DI)
> acf(resid(f7))
> pacf(resid(f7))
> AIC(f7, k=log(length(fitted(f7))))
```



As the plot shows, there was non-negligible residual autocorrelation for some of the lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1602.640.

Sixth experiment: Reduction in original dynamic model by dropping the DP, LDP, DI and LDI terms

The next step was to drop DP, or the difference between real price per capita, DI, or the difference between the per capita personal income, LDI, or the lagged difference for per capita personal income, and LDP, or the lagged difference for real price per gallon of gasoline and run the same test for the new reduced model:

```
> f8<-dyn$lm(Q~LQ+LDQ+P+I)
> acf(resid(f8))
> pacf(resid(f8))
> AIC(f8, k=log(length(fitted(f8))))
```

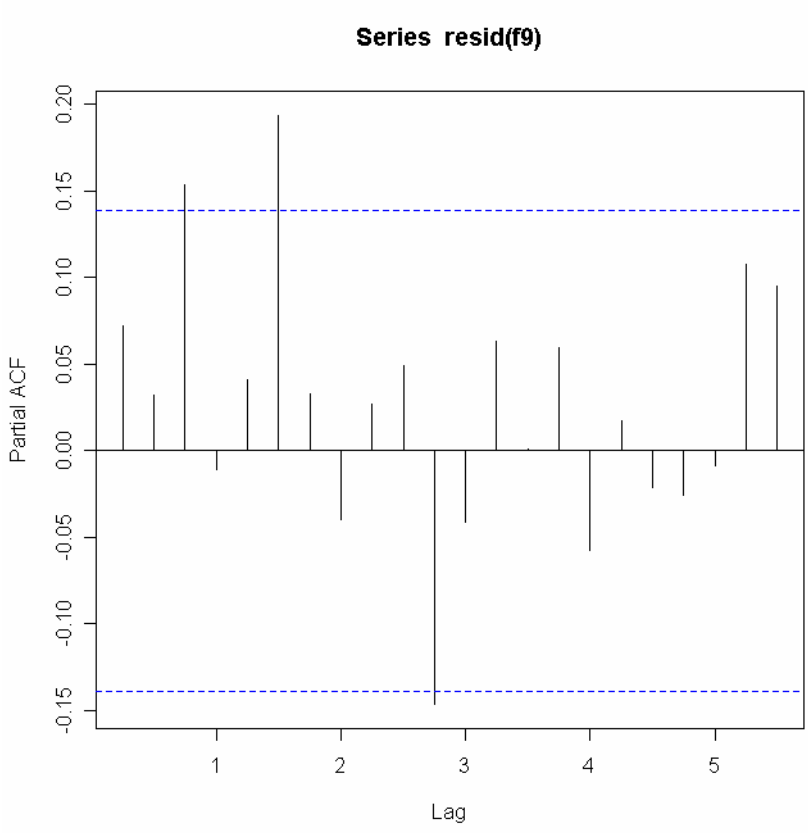
As the plot shows, there was non-negligible residual autocorrelation for some of the lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1218.523.

Seventh experiment: Reduction in original dynamic model by dropping the DP, LDP, LDI terms

The next step was to drop DP, or the difference between real price per capita, LDI, or the lagged difference for per capita personal income, and LDP, or the lagged difference for real price per gallon of gasoline and run the same test for the new reduced model:

```
> f9<-dyn$lm(Q~LQ+LDQ+P+I+DI)
> acf(resid(f9))
> pacf(resid(f9))
> AIC(f9, k=log(length(fitted(f9))))
```



As the plot shows, there was non-negligible residual autocorrelation for some of the

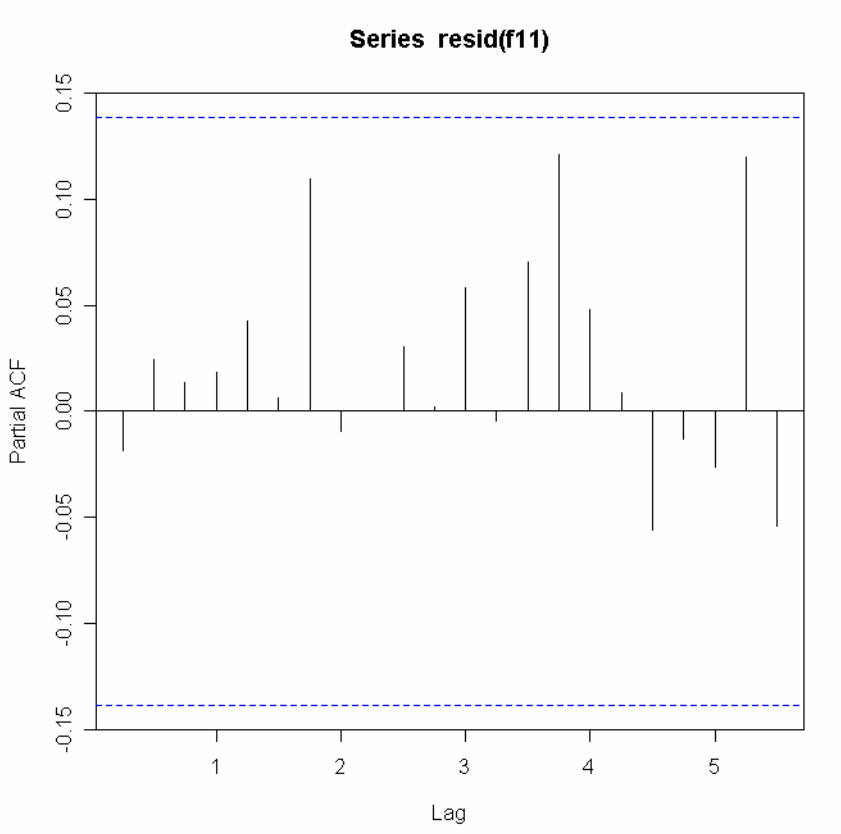
lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1218.264.

<u>Ninth experiment: Reduction in original dynamic model by dropping the DI, LDI terms</u>

The next step was to drop DI, or the difference between per capita personal income, LDI, or the lagged difference for per capita personal income, and run the same test for the new reduced model:

```
> f11<-dyn$lm(Q~LQ+LDQ+P+DP+LDP+I)
> pacf(resid(f11))
> acf(resid(f11))
> AIC(f11, k=log(length(fitted(f11))))
```



As the plot shows, there was no significant residual autocorrelation for any of the lagged values, and the BIC or SBC (Schwarz's Bayesian criterion) provided a value of:

-1588.962.